$$g(x) = -x, f(x) = x\sin\frac{1}{x}, h(x) = x$$

$$\lim_{x\to 0} g(x) = 0, \lim_{x\to 0} h(x) = 0$$

and have $\lim\limits_{x\to 0} x\sin\frac{1}{x} = 0.$

By Sandwich theorem

$$\lim_{x\to 0}\left(x\sin\frac{1}{x}\right) = 0 = h(0).$$

Therefore $h(x)$ is continuous in the entire real line.

Example 9.37

A tomato wholesaler finds that the price of a newly harvested tomatoes is ₹0.16 per kg if he purchases fewer than 100 kgs each day. However, if he purchases at least 100 kgs daily, the price drops to ₹0.14 per kg. Find the total cost function and discuss the cost when the purchase is 100 kgs.

Solution

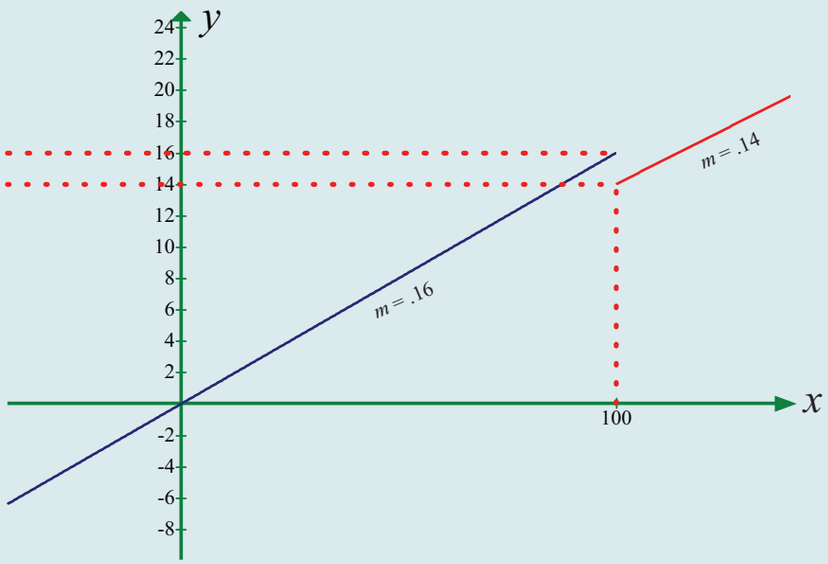


Fig.9.37

Let x denote the number of kilograms bought per day and C denote the cost. Then,

$$C(x) = \begin{cases} 0.16x, & \text{if } 0 \le x < 100 \\ 0.14x, & \text{if } x \ge 100 \end{cases}.$$

The sketch of this function is shown in Fig. 9.37.

It is discontinuous at $x = 100$ since $\lim\limits_{x\to 100^-} C(x) = 16$ and $\lim\limits_{x\to 100^+} C(x) = 14$.

Note that $C(100) = 14$. Thus, $\lim\limits_{x\to 100^-} C(x) = 16 \neq 14 = \lim\limits_{x\to 100^+} C(x) = C(100)$.

Note also that the function jumps from one finite value 14 to another finite value 16.

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